

Number Guessing Game — EMU8086 Assembly Language Project

Project Title

Number Guessing Game using Assembly Language (EMU8086)

Project Description

This project is a Number Guessing Game developed in Assembly Language using EMU8086. The game generates a random number between 0–9 and allows the player to guess the correct number within limited attempts.

The project demonstrates important Computer Organization and Assembly Language concepts including conditional jumps, loops, stack usage, arithmetic operations, input/output handling, random number generation, score tracking, and replay functionality.

The game provides interactive feedback to the user such as:

- Correct Guess
- Too High
- Too Low
- Game Over

The project also includes a score system and play-again feature to improve user interaction.

Features

- Random number generation
- User input handling
- Number comparison using CMP
- Conditional jumps (JE, JG, JL)
- Limited attempts system
- Score tracking
- Replay option
- DOS interrupt handling using INT 21H
- Simple and beginner-friendly Assembly Language logic

Technologies Used

- Assembly Language
- EMU8086 Emulator
- DOS Interrupts

Concepts Implemented

- CMP Instruction
- Conditional Jumps
- Loops
- Variables
- ASCII Conversion
- Random Number Logic
- Input/Output Handling
- Procedures and Labels

How the Game Works

1. The program generates a random number.
2. The user enters a guess between 0–9.
3. The program compares the guess with the generated number.
4. The program displays:
 - Correct!
 - Too High!
 - Too Low!
5. The player gets only 5 attempts.
6. If the player guesses correctly, score increases.
7. After completion, the player can choose to play again.

Sample Output

Guess Number (0-9): 5

Correct!

Score = 1

Play Again? (Y/N):

Learning Outcomes

Through this project, the following Assembly Language concepts were practiced:

- Arithmetic instructions
- Comparison instructions
- Conditional branching
- DOS interrupts
- User interaction
- Random value generation
- Program flow control

Conclusion

This project successfully demonstrates how Assembly Language can be used to create an interactive mini-game using EMU8086. It combines multiple low-level programming concepts into one practical project and helps strengthen understanding of computer organization and assembly programming.

THE END